ReBeat: X-point signed reference errors · Cognitive workload
n=22 participants; 433 strict common-reference beats · complete range; linear centre ±180 ms

Negative (early) · Positive (late) · Zero error

X1 · First local minimum
X2 · Global minimum
X3 · d3 upward zero
X4 · d3 minimum
X5 · 45% downstroke
X6 · 25% downstroke
X7 · d2 upward zero
X8 · d2 after minimum
X9 · d2 minimum
X10 · Weissler HR
X11 · Adaptive half descent
X12 · RR-adaptive minimum
X13 · 15% downstroke

Heartbeats (count)

Earlier ← signed detected − reference error (ms) → Later